

ITSS SOFTWARE DEVELOPMENT/ SOFTWARE DESIGN AND CONSTRUCTION

Lab 02: Use Case Specification

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1. SUBMISSION GUIDELINES

You are required to push all your work via the Assignment on Microsoft Teams. For this lab, you have to turn in:

- The Word files (in DOC or DOCX format) for the specification of three use cases: “Place Order”, “Pay Order” and “Pay Order by Credit Card”.

Deadline: Please refer to the assignment which was created by the lecturer (or the teaching assistant) on MS Teams.

This is an individual lab. Therefore, if we detect any sign of copying, both the one who copies and the one who lets others copy will receive a score of 0. If this happens more than twice, you will be banned from taking the final exam of this course.

2. USE CASE SPECIFICATION FOR “PLACE ORDER”

In the AIMS Project, use case “Place Order” describes the interaction between customers and AIMS software when the customer wishes to place order.

2.1. FLOW OF EVENTS

Naturally, we describe how a use case begins and ends in order to understand its purpose. We can consider a basic flow of events for use case “Place Order” as follows.

Please notice that: This is merely an example. You should carefully read the problem statement to determine this flow of events.

- Step 1.** Customer requests to place order
- Step 2.** AIMS software checks the availability of products in the cart
- Step 3.** AIMS software displays the form of delivery information with order information
- Step 4.** Customer enters and submits delivery information (see Table 1)
- Step 5.** AIMS software checks if the delivery information is valid
- Step 6.** AIMS software calculates shipping fee

Step 7. AIMS software displays invoice screen (see Table 2)

Step 8. **The AIMS software calls use case “Pay order”**

Step 9. The AIMS software saves a new order

Step 10. The AIMS software makes the cart empty

Step 11. The AIMS software sends email about the order notification and information

Step 12. The AIMS software displays the successful order notification, the order and the transaction information (see Table 3)

We specify what happens when, for each action. Remember this text *will be used to identify test cases*. However, things hardly ever go as planned. Taking the above flow as our initial basic flow, we would try to analyse and improve our flow of events for UC “Place Order”.

Question: Should we trust any input to an action/event?

It depends on the question or the request and the way it has been asked or responded. If the response must comply with a rule or a regulation, we must question the validity of the input. In case of humans, they tend to follow the given instructions in an incorrect manner even after several trials. In cases of non-human actors, there are a lot of factors that heavily affect the input from other systems, e.g., noise in transmission. Hence, we need to give them the chance to try repeatedly until their input is valid or within an acceptable limitation to be at least. However, there are cases in which we must take the input for granted since it is unnecessary or beyond the control of the system. To illustrate, in the form of delivery information, we should not check if the user has filled the real name of the receiver in the field for receiver name, yet we must not let the user leave it blank.

Question: What data is exchanged between actor and use case and between use case and use case?

In this use case, customer needs to provide with delivery information. Some or all attributes of these information may play critical roles in input validation, so we need to specify the attributes of the input. To illustrate, the input data of delivery information may include these data fields:

Table 1- Input data of delivery information

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Receiver Name		Yes		Do Minh Hieu
2.	Phone Number		Yes	- 10 digits - May contains separator "." and/or space and/or "-" in the middle of the phone number (one separator type), interleave with digits - Starts with "0"	0987 654 321 0987654321 0987.654.321 09.87.65.43.21 0987-654-321
3.	Province	Choose from a list	Yes		Hanoi
4.	Address		Yes		12, 34 Alley of Tran Thai Tong street, Cau Giay district
5.	Shipping instructions		No		

We also need to specify which the output to the actor(s) since it is the main factor that impacts on the input from actor(s). For instance, the output data are shown in the following table.

Table 2-Output data of invoice screen

No	Data fields	Description	Display format	Example
1.	Title	Title of a media product		DVD Phim Vượt ngục
2.	Price	Price of the corresponding media product	▪ Comma for thousands separator ▪ Positive integer ▪ Right alignment	123,000
3.	Quantity	Quantity of the corresponding media	▪ Positive integer ▪ Right alignment	2

No	Data fields	Description	Display format	Example
4.	Amount	Total money of the corresponding media	▪ Comma for thousands separator ▪ Positive integer ▪ Right alignment	246,000
5.	Subtotal	Total amount of all products in the order		2,316,600
6.	Shipping fee			30,000
7.	Total			2,346,600

Table 3-Output data of general information of order and transaction info

No	Data field	Description	Display format	Example
1.	Customer name			Do Minh Hieu
2.	Phone number			0987654321
3.	Province			Hanoi
4.	Address			12, 34 Alley of Tran Thai Tong street, Cau Giay district
5.	Total amount		Right alignment Vietnamese currency (VNĐ) Vietnamese locale	1.200.000 VNĐ
6.	Transaction ID			
7.	Transaction content			
8.	Transaction date		dd/mm/yyyy	05/10/2023

Note that we do not describe the details of the user interface unless it is necessary to understand the behavior of the system. Specifying user interface details too early will limit design options.

Now, we can finally validate the data. For list of media in the cart, we need to check if a media is out-of-stock. For delivery information, we need to check if a mandatory

field is left blank and valid condition for the phone number. Thus, we need insert at least two more events into the flow so as to validate the two corresponding inputs.

After validation, in case there is an exception, the flow cannot continue normally. Consequently, we need alternative flows or sub-flows for the next events in these cases. For instances, the sub-flows for use case “Place Order” is shown as follows (may be not complete) in Table 4.

Table 4 - Alternative flow of events for UC “Place Order”

No	Location	Condition	Action	Resume location
1.	At Step 3	If the products are not available	The AIMS software notifies that the products in the cart are not available and stay at the use case “View cart”	Use case ends
2.	At Step 5	If the delivery info is invalid	AIMS software notifies that the delivery info is invalid (blank or wrong format)	At Step 3
3.	At Step 8	If the order payment is not successful or goes back from payment		At Step 5

The last questions are what we should save and when we save it.

By saving the data, we can save a lot of time and efforts for us, the system, and the users. To illustrate, the customer cannot finish placing order for some reasons. Thus, we can save some information for later such as the list of media in cart, so that the customer does not have to add them to the cart again.

2.2. USE CASE SPECIFICATION

Finally, we may provide the pre-condition and the post-condition. For example, the pre-condition for use case “Place Order” can be “There is at least one item in the cart.” A post-condition can be “A new order is created, and its information is sent via email to the customer or nothing happens if payment is not successful”.

Template of Use Case Specification is shown as below (see page 6).

Use Case “Name of use case”

1. Use case code

UC00X

2. Brief Description

This use case describes the interaction between <actor(s)> and <name_of_the_system> when <actor(s)> wish(es) to ...

3. Actors

3.1 Name of Actor 1

4. Preconditions

5. Basic Flow of Events

1. Actor ...
2. Software displays ... (see Table T).
3. ...

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step S	If ...	▪ Action 1	Resumes at Step Q
2.	At Step O	If ...	▪ Action 2	Use case ends

7. Input data

Table A-Input data of ...

No	Data fields	Description	Mandatory	Valid condition	Example
1.					

8. Output data

Table B-Output data of ...

No	Data fields	Description	Display format	Example
1.				

9. Postconditions

For this part, following the above suggestion, you are asked to make a use case specification for use case “Place Order” by using the template. Remember to validate data and save information if need be. When you finish the task of this part, please name your file as the convention “<StudentID>_<StudentName>_PlaceOrder” (for example, 20221234_NguyenHaiAnh_PlaceOrder) for submission.

3. USE CASE SPECIFICATION FOR “PAY ORDER”

In this part, you are asked to fill in the provided template for the use case specification for use case “Pay Order”. When you finish the task of this part, please name your file as the convention “<StudentID>_<StudentName>_PayOrder” (for example, 20221234_NguyenHaiAnh_PayOrder) for submission.

In AIMS, by default, the customer will pay the order via the QR code on the payment screen, which was created by calling the API of VietQR. This use case describes the interactions between the AIMS software with the customer and VietQR when the customer desires to pay order. An example for the basic flow of events is listed as follows:

- Step 1.** Customer asks to pay the order
- Step 2.** AIMS software calls the API of VietQR
- Step 3.** VietQR generates the QR code
- Step 4.** AIMS software displays the QR code on the payment screen together with the invoice information
- Step 5.** Customer scans the QR code to complete the transfer in their banking application
- Step 6.** AIMS software verifies the transaction result (via bank API or callback from payment partner)
- Step 7.** AIMS software updates the invoice status

The alternative flows of events of this use case are illustrated in the following table.

No	Location	Condition	Action	Resume location
1.	At Step 5	If the customer choose to pay order by credit card	AIMS Software calls the use case “ Pay order by credit card ”	Use case ends

No	Location	Condition	Action	Resume location
2.	At Step 6	If the transaction is failed		Use case ends

4. USE CASE SPECIFICATION FOR “PAY ORDER BY CREDIT CARD”

You are asked to update the use case specification for use case “Pay Order” and/or to make another use case specification by using the provided template (e.g., you can model the relationship between use case “Pay Order by Credit Card” and use case “Pay Order” as an extension) for the additional use case “Pay Order by Credit Card”.

When you finish the task of this part, please name your file following the convention “<StudentID>_<StudentName>_PayOrderByCreditCard” for submission.

Please notice that: If you consider the use case “Pay Order by Credit Card” in the use case “Pay Order”, you don’t need to complete this task. However, remember to include flow of events and other information of this use case inside the use case “Pay Order”.